DATA ANALYTICS PROJECT

BlinkIT Grocery Sales Analysis

Retail & Grocery Sales Analytics

Prepared for: Sapphire IQ

Tools: Python, Pandas, NumPy, Matplotlib, Seaborn and Jupyter Notebook

Dataset: BlinkIT Grocery Data

Project purpose

Analyze grocery sales records across product categories, outlet formats, outlet locations and product characteristics. The goal is to summarize performance patterns and present practical, evidence-based questions and recommendations for business review.

Prepared from the supplied cleaned dataset and project notebook. Sales are reported in the dataset's recorded units; no currency is assumed.

1. Executive Summary

The cleaned dataset contains **8,523 records**, **1,559 unique product identifiers**, **10** outlets and **16** product categories. Total recorded sales are **1,201,681.49** dataset units, average sales per record are **140.99**, and the mean product rating is **3.97**.

The largest category sales total is Fruits and Vegetables, closely followed by Snack Foods. Supermarket Type 1 contributes the largest share among outlet formats, and Tier 3 contributes the largest share among location tiers. The correlation between item visibility and sales is close to zero in this dataset; this is a descriptive linear association and does not establish causation.

Metric	Result
Records	8,523
Total sales (dataset units)	1,201,681.49
Average sales per record	140.99
Average rating	3.97 / 5
Unique products	1,559
Outlets	10
Product categories	16

2. Project Objectives

• Understand the business and its sales performance across products, categories, outlets and locations.

• Assess data reliability by checking missing values, duplicate records, unusual values and inconsistent labels.

• Explore factors and segments associated with stronger or weaker sales performance.

• Summarize findings and provide practical recommendations to support management decisions.

3. Dataset Overview

The analysis uses the supplied BlinkIT grocery dataset. The cleaned file contains 8,523 rows and 13 columns. The fields cover item fat content, item identifier, item type, outlet establishment year, outlet identifier, outlet location tier, outlet size, outlet type, item visibility, item weight, sales, rating and a derived weight category.

4. Data Cleaning & Preparation

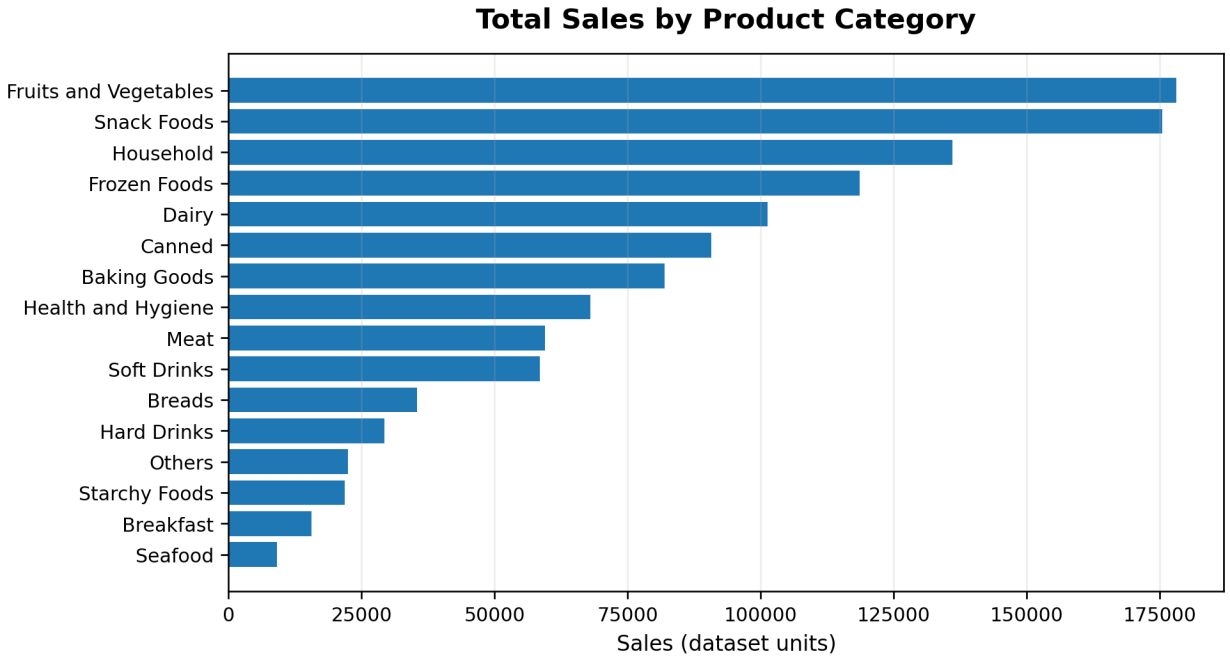
The notebook creates a working copy of the source data, assesses missing values and duplicates, imputes missing item weights using product-level information and then category-level medians for remaining gaps, standardizes Item Fat Content labels, checks duplicate rows, flags potential outliers using the IQR method, validates the cleaned data and exports a CSV.

Validation of the supplied cleaned CSV shows **0** missing values and **0** duplicate rows. The final cleaned file has 8,523 rows and 13 columns.

Outlier flags are not automatically removed. Extreme values may represent genuine sales or product observations and should be investigated in context.

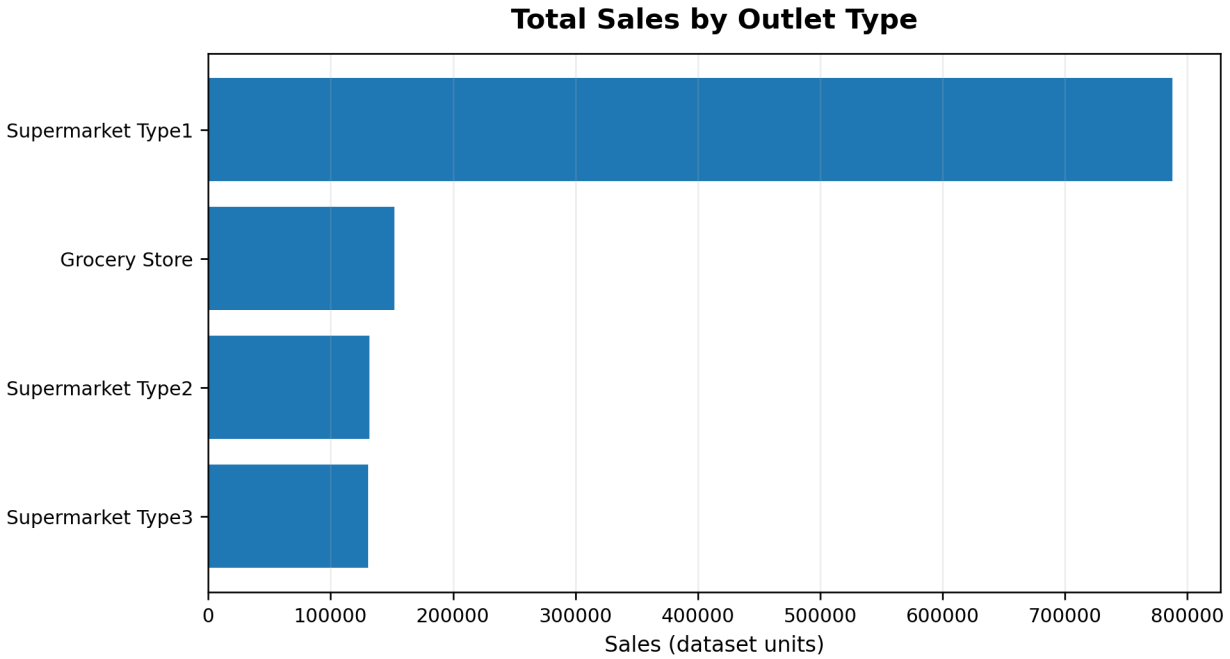
5. Exploratory Data Analysis

5.1 Sales by Product Category



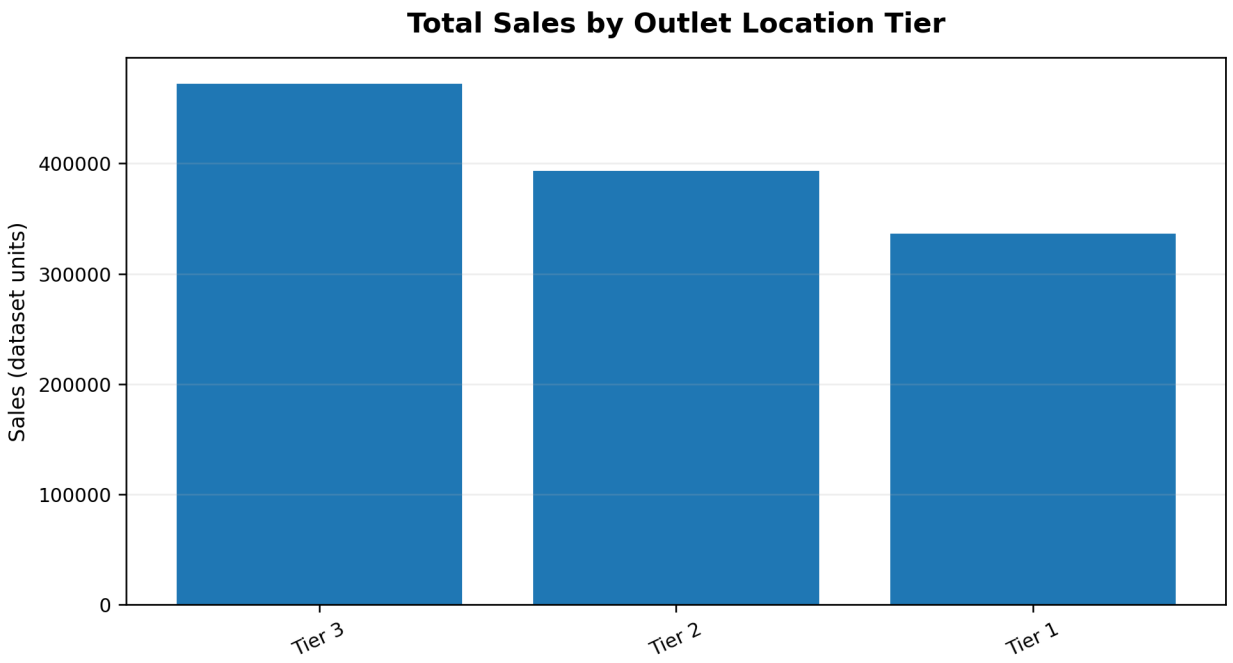
Fruits and Vegetables lead category sales at 178,124.08, followed by Snack Foods at 175,433.92. These are the highest total-contributing categories in the supplied data. Total sales can be influenced by the number of records in each category, so average sales and product mix should also be considered.

5.2 Sales by Outlet Type



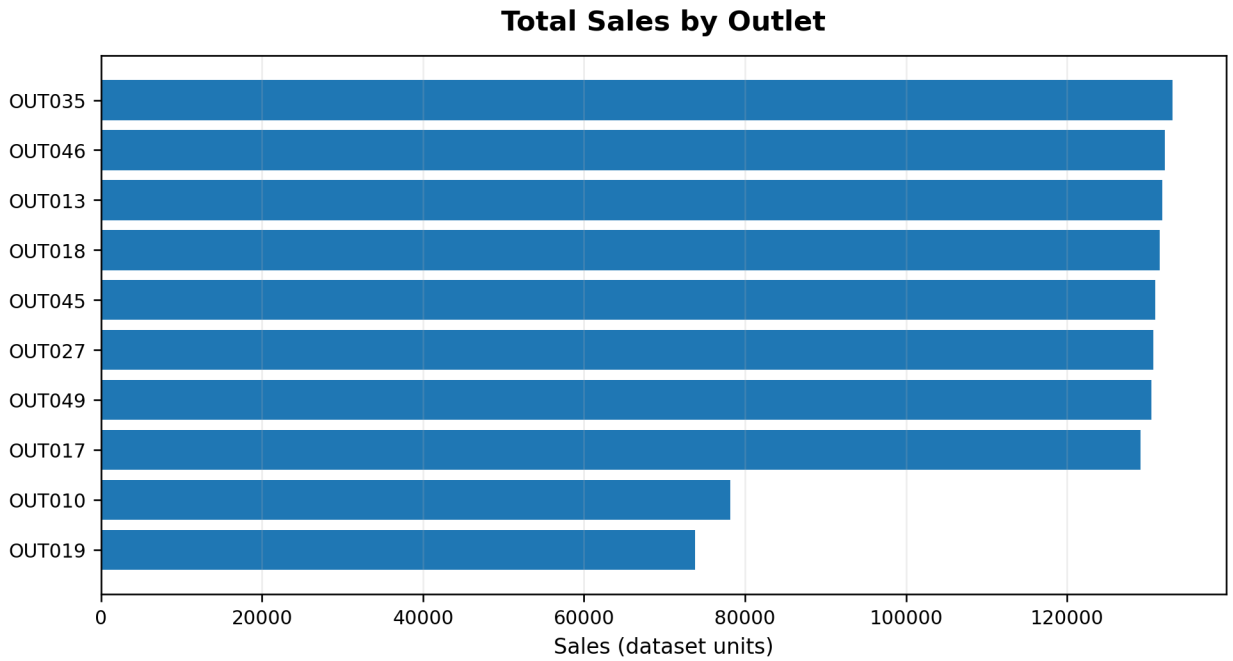
Supermarket Type1 records the largest total sales, 787,549.89, representing 65.54% of total sales. This identifies its contribution in this dataset; it does not by itself measure profitability or operational efficiency.

5.3 Sales by Location Tier



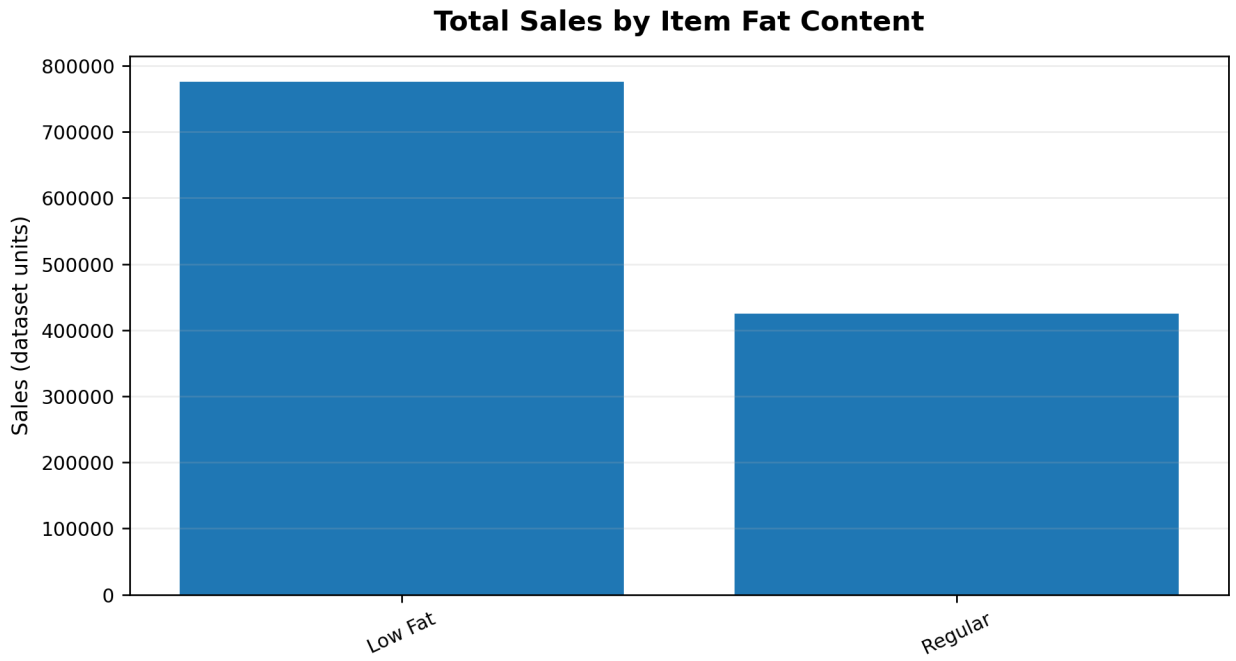
Tier 3 contributes the largest location-tier sales total (472,133.03; 39.29% of total). Differences in outlet count, format and product mix may influence this comparison.

5.4 Sales by Outlet

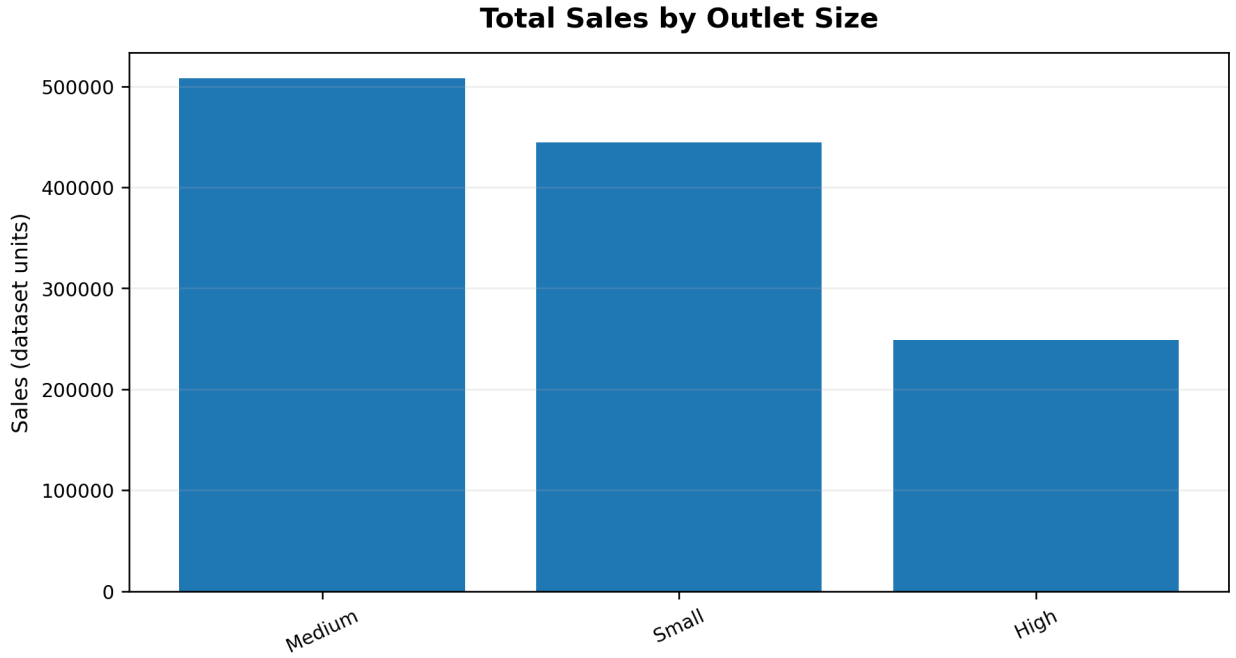


OUT035 is the highest outlet by total sales (133,103.91). The lowest is OUT019 (73,807.58). Outlet-level comparisons should be interpreted alongside outlet type, size, location and record counts.

5.5 Sales by Fat Content and Outlet Size



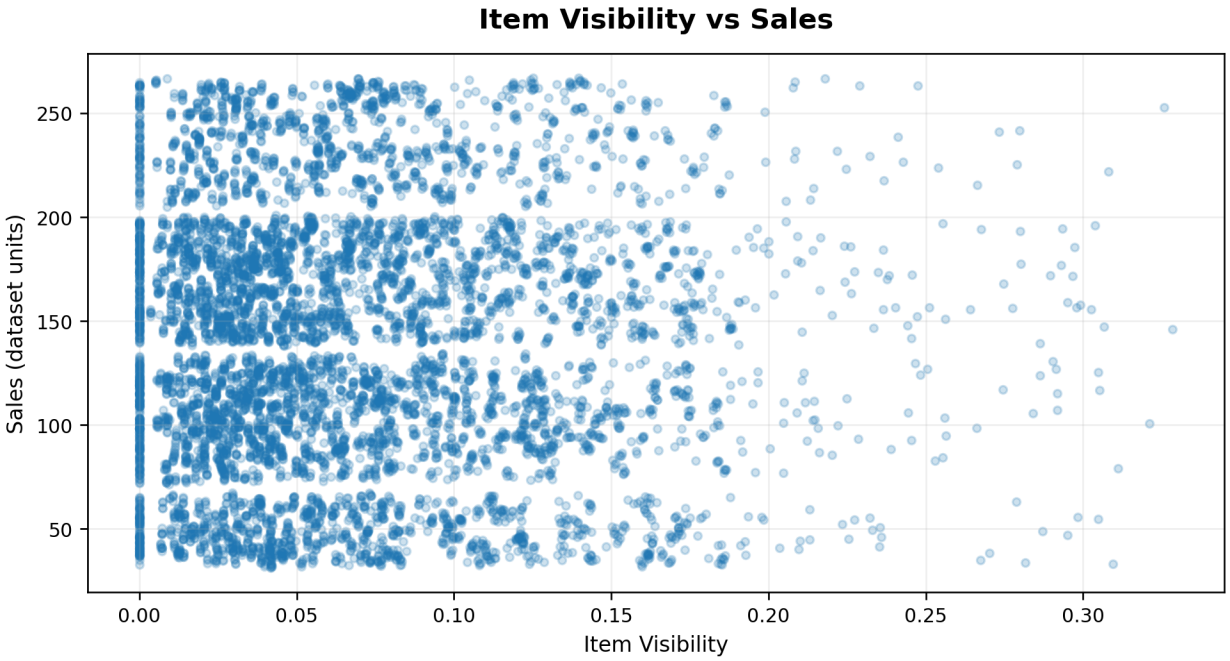
Low Fat products contribute 64.60% of recorded sales; Regular products contribute 35.40%. This describes sales mix and should not be interpreted as a causal preference without additional customer or demand evidence.



Medium outlets have the highest total sales by outlet-size grouping (507,895.74). Total group sales can reflect how many records/outlets fall into each group.

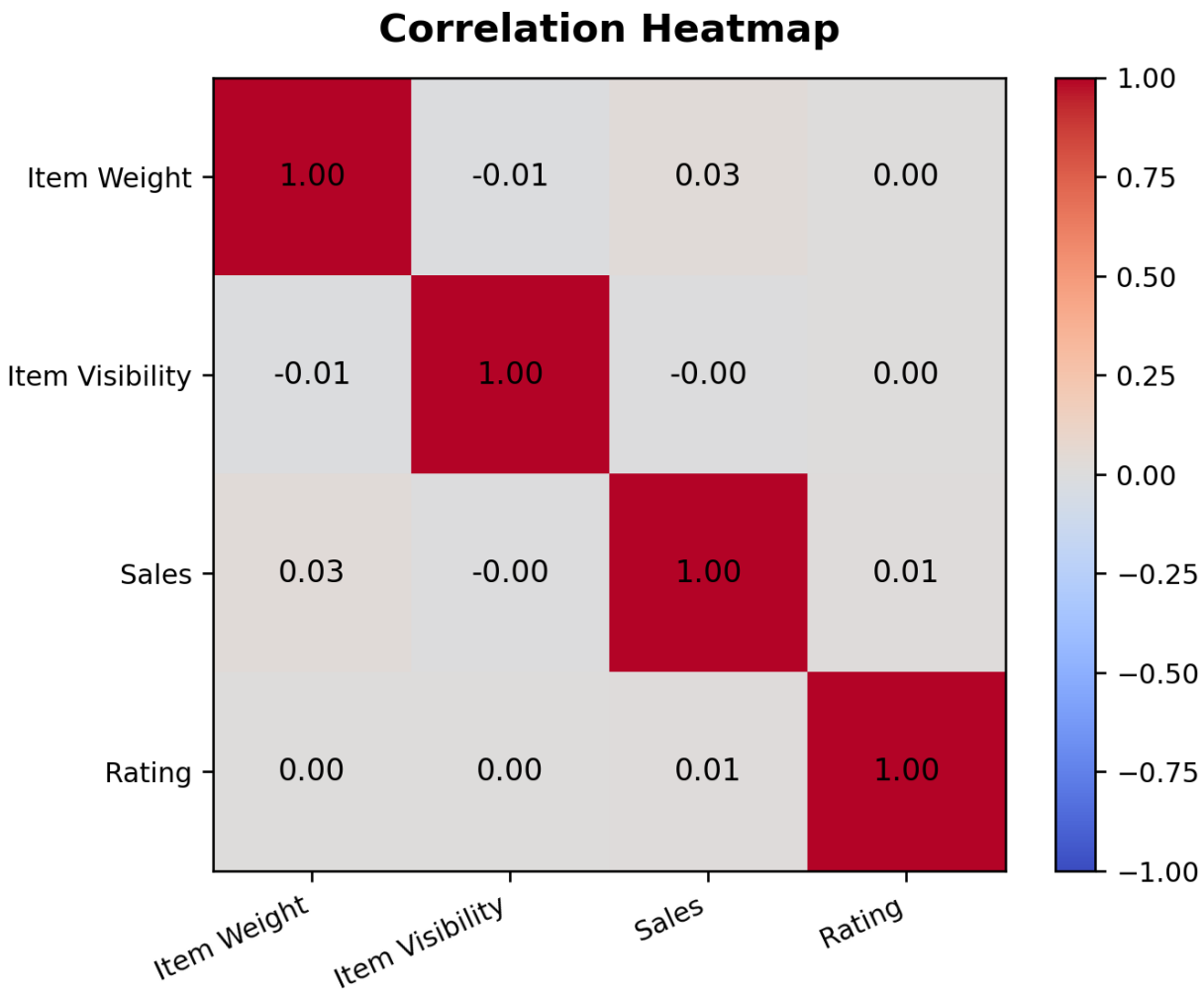
6. Relationships and Advanced Analysis

6.1 Item Visibility and Sales



The Pearson correlation between Item Visibility and Sales is -0.0013, indicating virtually no linear association in this dataset. Correlation does not establish causation, and a near-zero linear correlation does not rule out nonlinear or segment-specific relationships.

6.2 Numerical Correlation Heatmap



The correlation between Item Weight and Sales is 0.0260, also a very weak positive linear association. The heatmap is descriptive; product category, price and outlet context may affect observed sales.

6.3 Outlet Performance Snapshot

Group	Outlet	Total sales (dataset units)
Top 3	OUT035	133,103.91
Top 3	OUT046	132,113.37
Top 3	OUT013	131,809.02
Bottom 3	OUT019	73,807.58
Bottom 3	OUT010	78,131.57
Bottom 3	OUT017	129,103.96

7. Key Findings

• Fruits and Vegetables are the top category by total sales (178,124.08), with Snack Foods second (175,433.92).

• Supermarket Type 1 is the largest outlet-format contributor (65.54% of total sales).

• Tier 3 locations contribute the largest share by location tier (39.29%).

• Low Fat products account for 64.60% of recorded sales.

• Item Visibility has a near-zero Pearson correlation with Sales (-0.0013) in the full dataset.

• OUTLET OUT035 is the highest outlet by total sales; this comparison should be contextualized using outlet characteristics.

8. Business Recommendations

• **Inventory planning:** Review replenishment levels for leading categories such as Fruits and Vegetables and Snack Foods, while monitoring stock-outs and wastage.

• **Outlet benchmarking:** Compare top- and lower-sales outlets using outlet format, size, location, record count and average sales before recommending operational changes.

• **Location-specific assortment:** Examine category mix by location tier and tailor assortment only where repeated demand evidence supports it.

• **Balanced product mix:** Maintain availability across both Low Fat and Regular products and monitor changes over time rather than relying on one aggregate comparison.

• **Test visibility interventions:** Since the overall linear correlation is near zero, use controlled placement tests or more detailed segment analysis before investing in visibility changes.

• **Improve future analysis:** Add time-period, price, quantity, margin, inventory and promotion data if available to distinguish revenue, volume, profitability and demand drivers.

9. Limitations

• The supplied data is a snapshot and does not provide a time series suitable for trend or seasonality analysis.

• The Sales field is reported in the dataset's recorded units; the supplied files do not establish a currency or whether the measure is revenue, gross sales or another sales metric.

• The dataset does not include explicit cost, profit, quantity, stock-out, promotion or customer-level information.

• Aggregated comparisons and correlations are descriptive and should not be interpreted as causal effects.

10. Conclusion

This project organizes and cleans the supplied grocery sales data, explores product and outlet performance, compares sales across business dimensions and summarizes evidence-based recommendations. The results provide a useful descriptive view of the dataset and a starting point for further operational investigation. Management decisions should be validated with additional context such as profitability, inventory availability, sales volume and time-based performance.

Appendix: Files and Tools

Analysis tools: Python, Pandas, NumPy, Matplotlib, Seaborn and Jupyter Notebook.
Primary analysis artifact: BlinkIT_Grocery_Sales_Analysis_Polished.ipynb
Cleaned data artifact: BlinkIT_Cleaned_Data.csv